HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and extended a production **Python AI service**, integrating with **LLM providers** for real-time intelligent response streaming.
- Implemented **LLM tool calling** and agentic loops, including detailed **prompt engineering** for complex workflows.
- Designed and executed **LLM evaluations** using Python, tracking key quality metrics to ensure feature evolution stability.
- Wrote **SQL queries** against **analytical databases** to ground AI responses with factual data, ensuring contextually relevant outputs.
- Managed **LLM cost and usage observability** through custom Python scripts and monitoring tools, identifying optimization opportunities.
- Implemented **service-to-service authentication** patterns and **PII handling** protocols to maintain data security and compliance.
- Containerized the AI service using **Docker** and deployed it to a scalable **AWS environment**, ensuring high availability.

SKILLS

Languages – Python, SQL, Ruby

AI/LLM – LLM APIs, Tool Use, Prompt Engineering, Agentic Loops, LLM Evaluations, Generative AI, RAG

Databases – ClickHouse, PostgreSQL, Analytical Databases

Cloud/DevOps – AWS, Docker, CI/CD, Observability

Frameworks/Libraries – LangChain, Pandas, NumPy, Scikit-learn

Protocols – REST APIs, Streaming HTTP

Security – Prompt Injection, PII Handling, Service-to-Service Auth

PROJECTS

AI-Powered Workforce Analytics Service

- Built a standalone **Python service** to process workforce data and integrate with **LLM APIs** for generating insights.
- Developed **prompt engineering** strategies to guide LLMs in analyzing scheduling and attendance data, grounding responses with **SQL queries** against an analytical store.
- Implemented **streaming HTTP** responses for real-time delivery of AI-generated workforce recommendations to end-users.

LLM Evaluation Framework for Quality Assurance

- Created a Python framework to systematically evaluate the quality of LLM outputs for workforce management use cases.
- Designed and implemented metrics for evaluating factual accuracy, relevance, and safety of generated responses, tracking performance across model versions.
- Integrated the framework with CI/CD pipelines to automate checks before deploying new LLM features.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)